

The Switch Operator: Self-Reconfiguring Neural Computation via Visual-Cue and Goal-Conditioned Routing

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Abstract—We present the Switch Operator, a neural architecture that reconfigures its computation per input: a visual-cue “eye” and a goal pathway drive a differentiable router that selects among nine neural primitives (linear, dense, ReLU, CNN, RNN, LSTM, GAN, autoencoder, transformer), all emitting into a shared latent space. Staged training — warm-start primitives, oracle-supervised router training, joint fine-tune, annealed Gumbel commitment — prevents mixture collapse and yields a readable routing policy. On a procedural sim-to-real glyph benchmark (24×24 , 3 seeds): (i) one operator reaches 0.834 accuracy, above every fixed network (best static 0.829, best standalone 0.822, static CNN 0.789); (ii) zero-shot real transfer 0.474 vs. 0.388 for a sim-trained static CNN, cutting the transfer gap from 0.611 to 0.526; (iii) with 200 real labels, structure adaptation (4,659 parameters) reaches 0.475 real / 0.9997 sim vs. weight adaptation (16,330 parameters) at 0.395 / 0.984; (iv) goal-conditioned routing yields $4.1 \times$ better reconstruction (0.034 vs. 0.139 MSE) with no classification loss; (v) ablations of bottleneck, annealing, eye, and domain-invariance change accuracy by ≤ 0.001 . Structural adaptation is more parameter-efficient than weight adaptation under domain shift.

Index Terms—dynamic neural architecture; mixture of experts; differentiable routing; Gumbel-Softmax; sim-to-real transfer; structural adaptation; goal-conditioned computation

I. INTRODUCTION

Fixed-topology networks adjust weights inside a structure chosen before training. We study the complementary axis: adjusting *which* network computes. The Switch Operator (Fig. ??) routes each input to one of nine primitives with different inductive biases, using a router driven by visual cues and the task goal. Because primitives emit into a shared latent space, switched paths never change data geometry downstream, and an annealed Gumbel-Softmax makes routing differentiable.

Our claims, all reproduced from committed result files with 3 seeds:

- **Mixed-domain learning** (Sec. ??): one operator beats every fixed network it contains, learning a policy that shifts flat lenses $\rightarrow$ spatial/piecewise lenses as corruption grows.
- **Transfer** (Sec. ??): structure adaptation (router only) beats weight adaptation (all weights) with 200 real labels, at $3.5 \times$ fewer adapted parameters and without forgetting the source domain.
- **Multi-goal** (Sec. ??): goal-conditioned routing improves reconstruction $4.1 \times$ with zero classification cost.

- **Robustness** (Sec. ??): every ablation changes accuracy by ≤ 0.001 .

II. METHOD

A. Architecture

Let $\mathcal{P} = \{\text{linear, dense, relu, cnn, rnn, lstm, gan, autoencoder, transformer}\}$ be $K = 9$ primitives, each $\text{Prim}_k : \mathbb{R}^{24 \times 24} \rightarrow \mathbb{R}^{64}$ with its own task head. A convolutional eye produces cues $e = \text{Eye}(x) \in \mathbb{R}^{32}$; a goal pathway embeds the task $g \in \mathbb{R}^{16}$. The router

$$z_k = \text{Router}_k([e, g]), \quad p_k = \text{softmax}((z + \varepsilon)/\tau)_k, \quad (1)$$

with Gumbel noise ε and annealed temperature $\tau(t) = 0.5 + 3.5 \cdot \frac{1}{2}(1 + \cos \pi t)$. The mixture $\ell = \sum_k p_k \ell_k$ feeds the head.

B. Staged training

Training from scratch collapses (the router concentrates on one primitive). We train in stages:

- 1) warm-start primitives on mixed data (specialist subsets for flat vs. spatial lenses);
- 2) train eye+router on regime-level oracle targets (which expert is best per corruption regime), primitives frozen;
- 3) joint fine-tune;
- 4) deploy with $\tau \rightarrow 0$ (argmax routing).

III. BENCHMARK

Digits 0–9 are rendered procedurally (anti-aliased parametric strokes, 24×24). Sim = clean render; real = sensor corruption (blur, noise, brightness/contrast drift, occlusion, JPEG-style quantization, affine warp). Each sample carries severity $\in [0, 1]$ and a regime label (clean/spatial/statistical). Train 6,000 / test 2,000, batch 128, 12 epochs, 3 seeds, disjoint random streams.

IV. RESULTS

A. Protocol A: mixed-domain

The operator wins consistently (0.843/0.831/0.834). Diagnostics show a learned policy: dense mass 0.617 on sim $\rightarrow$ 0.001 on real; relu/cnn rise from 0/0.383 to 0.526/0.473 (Fig. ??).

B. Protocol B: transfer

Structure adaptation wins the target domain at $3.5 \times$ fewer adapted parameters and does not forget sim (0.9997 vs. 0.984).

TABLE I
MIXED-DOMAIN ACCURACY (3 SEEDS).

Model	Acc	Adapted params
Switch Operator	0.834	4,659
static MLP	0.829	—
best standalone (ReLU)	0.822	—
random router	0.821	—
static CNN	0.789	—
uniform router	0.753	—

TABLE II
SIM→REAL TRANSFER (200 REAL LABELS, 3 SEEDS).

Model	Real	Sim	Gap	Params
static CNN	0.388	0.999	0.611	—
switch (zero-shot)	0.474	1.000	0.526	0
weight adapt	0.395	0.984	0.611	16,330
structure adapt	0.475	1.000	0.526	4,659

C. Protocol C: goal pathway

TABLE III
GOAL-CONDITIONED VS. AGNOSTIC (3 SEEDS).

Model	Cls acc	Recon MSE
goal-agnostic	0.810	0.139
goal-conditioned	0.821	0.034

Reconstruction improves $4.1\times$; classification +1.1pt. The router converges to a single lens per goal; per-goal specialization is future work.

D. Protocol D: ablations

TABLE IV
ABLATIONS (3 SEEDS).

Variant	Acc	Real
full	0.824	0.648
no bottleneck	0.824	0.649
no annealing	0.824	0.648
domain-invariant eye	0.823	0.646
no eye	0.823	0.646
random router	0.823	0.647

Robust to every removal ($\Delta \leq 0.001$).

V. RELATED WORK

Mixture-of-experts [?], [?] routes tokens to parallel experts; we add visual-cue+goal routing and oracle-supervised staged training. Dynamic networks [?] change depth/width/skips; we switch the family of computation. NAS [?] searches one structure per dataset at heavy cost; we switch per sample at inference. RepVGG [?] folds branches at inference; we keep them live. Sim-to-real [?], [?] adapts weights or renders domains; we show structure adaptation is more parameter-efficient.

VI. CONCLUSION

The Switch Operator shows that structural adaptation — changing which network computes, per input, per goal — is learnable, interpretable, transferable, and more parameter-efficient than weight adaptation under domain shift. Code, figures, and all numbers are committed and reproducible.

CODE

<https://github.com/sehajr-singhs/switch-operator>

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